Glacial landscape evolution of the Patagonian Icefields

By L.M. Chen, University of Amsterdam, 2009

Introduction

Landscape evolution under glaciation is characterized by processes that are dynamically dependent on both environmental factors and each other. Glacial erosion, isostatic uplift responses, ELA positioning and derived mass balance, ice thickness and flow speed, relief and sedimentary landform production and removal are processes that although significant in their impact on glacial geomorphology remain incompletely understood, pertaining to both their dynamics as the spatial and temporal scales at which they operate (Stroeven and Swift, 2008). Presented in this paper are two case studies where mechanisms of non-climatic influence on ice dynamics have been proposed to solve current uncertainties in the reconstruction of the glacial landscape evolution history of the Patagonian icefields, to serve as an example of progress in this field and the methods by which it is achieved.

The Patagonian icefields

The Northern Patagonian icefield, which covers and area of 4200 km^2, is situated in Chile, South-America (Aniya, 1988). Its existence in temperate latitudes (47°S) is sustained by abundant precipitation over the icefield (2 to 11m of water equivalent per yr) generated as the Southern Westerlies are forced over the Andes (Casassa et al., 2002). It is a very climate-sensitive icefield (Kerr and Sugden, 1994; Rignot et al., 2003), which makes it an ideal subject area for attempts to distinguish between climatic and non-climatic icesheet dynamics.

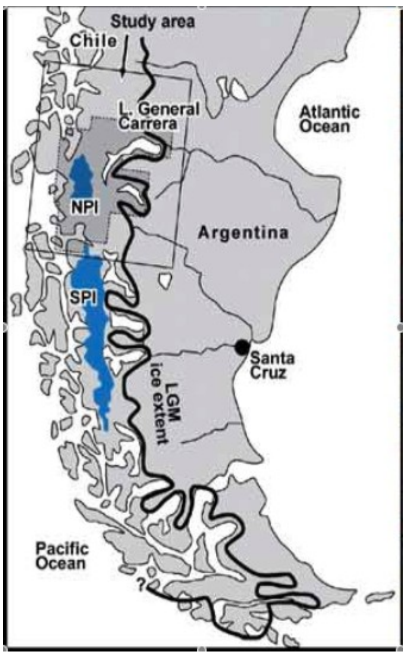

Figure 1: Northern Patagonian Icefield and the LGM extent (Glasser and Jansson, 2005)

The greater study area of Patagonian icefields is chosen between 38° and 54°S and at the Last Glacial Maximum or LGM, around (~800kyr) covered an area of about 422.000 km^2. Due to this regions glacial moraine records being one of the best-dated, most extensive on earth (Kaplan et al., 2009), it in turn provides an ideal subject area for studies in spatial and temporal ice dynamics.

In the study of the history of landscape development and glacial dynamics in these regions, problems have been encountered when numerical models using existing theory were unable to explain the geomorphological evidence observed in either area (Wenzens, 2002; Kaplan et al., 2009 respectively). Efforts were undertaken to propose new suggestions to improve these models, which will be discussed below.

Inference of dynamics by landform interpretation

In a study published in 2005, glacial landforms around the Northern Patagonian icefield have been mapped with various kinds of remote sensing data and interpreted on spatial continuity or resemblances to plausible patterns to yield a generalized map, a sample of which is provided in figure 2. Lineations were combined into flow-sets which were suggested to represent the former flow-lines of the icefield, and all major flow-sets were shown to be characterized by a narrow channel with sharp lateral boundaries, parallel conformity between lineations and, on the eastern side of the icefield where glaciers did not generally terminate in tidewater, lobate terminal moraines (Glasser and Jansson, 2005).

Existing models of the Northern Patagonian ice-sheet during the LGM have been unable to correctly represent the thickness of the ice and the location of the ice-divide (Hulton et al., 2002) when compared to geomorphological data, such as the presence of lineations at the location of the estimated ice-divide which conflicts with the suggestion by Boulton and Clark (1990) that non-erosive conditions prevail under ice-divides.

From the mapping culminating in the figure below, Glasser and Jansson suggested that the flow-sets represent fast flowing outlet glaciers, who as extensions of icestreams further upglacier would have strongly influenced the ice discharge patterns and mass configuration and therefore represented a partial decoupling of icesheet dynamics from climatically induced changes. This would imply a greater and more sensitive discharge of ice, lesser overall ice thickness and an ice divide further to the west. Numerical models modified to include this suggestion, although not yet run, might be able to concur with the mapped landforms and could provide an explanation of the prevailing dynamics of the icesheet and regional glacial landscape evolution during the LGM.

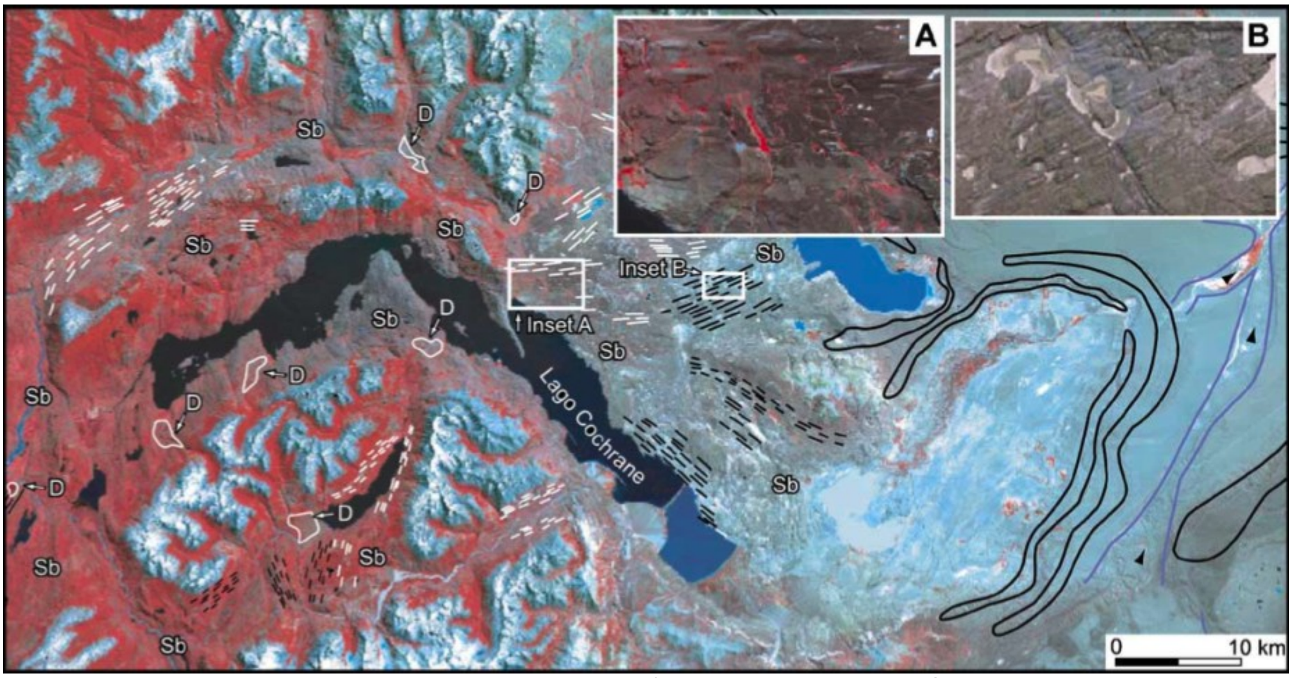

Figure 2: remote-sensing data with interpreted glacial landforms: scoured bedrock (Sb), glacial lineations (white and black lines), ice-contact deposits and deltas (D), terminal moraines (lobate features) and meltwater channels (blue lines). Insets show glacial lineations in sediment (A) and bedrock (B) (Glasser and Jansson, 2005).

Glacial erosion and a self-defeating mechanism

Another problem that needs resolving is the discrepancy between the glacial extent at the period of Greatest Patagonian Glaciation or GPG and that at the LGM.

According to Gibbons et al. (1984), the expected preservation for any random number of glacial successions is 2-3 moraines, since subsequent advances remove previously deposited moraines. For more then 1500km along the southern Andes a trend of more then three moraines persists, irrespective of tectonic or climatic setting and pointing to a non-random explanation (Kaplan et al., 2009). By mapping and dating the moraine records around Patagonia it has become clear that the extent of glaciation has generally been decreasing since the GPG, around 1.1 Ma (Caldenius, 1932, Rabassa and Clapperton, 1990; Mercer, 1983; Meglioli, 1992; Singer et al., 2004a; Coronato et al., 2004a; Rabassa et al., 2005), as opposed to the LGM at 800 kyr when the highest volume of ice occurred. An explanation for this discrepancy and the inability of existing numerical models to account for it has been sought in non-climatic effects such as the effect of glacial erosion on glacier dynamics.

In offering explanations of system dynamics in Antarctica, Clapperton (1990) proposed that under tectonic subsidence, valley-deepening shortened the distance over which a glacier had to transport its ice to effectively discharge it. How exactly an icemass would respond and possibly adjust is unclear; however the idea formed the basis of a hypotheses by Kaplan et al. who modified the concept.

By using known data in a somewhat simplified model, they simulated a situation where by adjusting the ELA, an icesheet with an extent resembling the extent prevalent during the LGM could self-sustain on current topography. Then, by infilling the topography below 500m with 30% and 100% of its elevation difference from that contour respectively two new glacial extents were calculated. They showed that the icesheet became less extensive under the new conditions. Although this conclusion does not concur with the landform data, where pre-LGM topography carried a greater glacial extent, the notable exclusion of infilling above the 500m line lead Kaplan et al. to form their suggestion: for the ice extent to have decreased, the accumulation area of a glacier must have changed, possibly leading to a less positive mass balance over time.

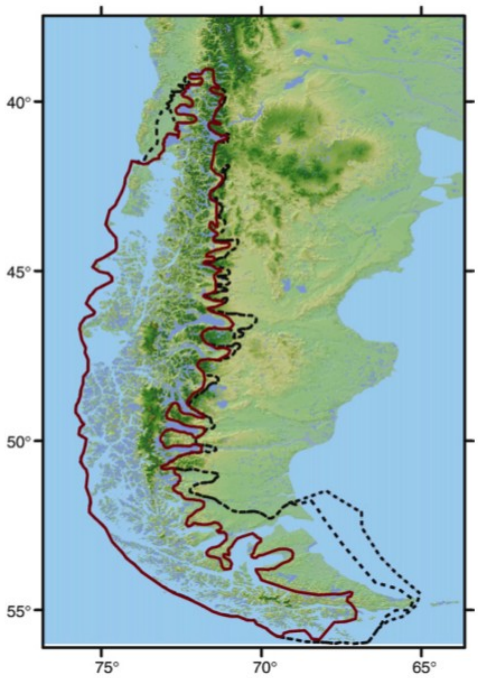

Figure 3: GPG extent (red) and LGM extent (dashed)

Resembling the self-defeating mechanism such as the ones proposed by Macgregor (2000) explaining similar extent reductions, the same concept in reverse by Brozovic et al. (1997) and Whipple et al. (1999), who proposed that glacial erosion will be a limiting factor to mountain height in general, or the idea from Sugden and John (1976) who proposed that areal scouring could result in a loss of accumulation area, the suggestion is an example of applying theoretic analogy and field data checked modeling. Although in reasoning Kaplan et al. proceed from an incomplete model that fails to account for the observed phenomenon to reversing the logic order into the field of undetermined variables and simplified modeling and deriving conclusions therefrom, the suggestion nonetheless dovetails with similar ideas and might be an apt solution to the current inconsistency.

Conclusions

Through inference from observation, theoretic analogy and numerical model testing, steps are being made to further our understanding of icesheet dynamics. Fields of incomplete understanding such as above-glacier valley wall retreat, supraglacial till with its radiation shielding properties, subglacial erosion and deposition dynamics, topographic control of glacial erosion and the influence of lithological strength are all still under debate and subject for further study in these or other ways. For both our comprehension of paleoglaciological landscape evolution and the critical test it provides for global climate models, and the methods with which to properly infer such system dynamics, these studies could prove to be essential.

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